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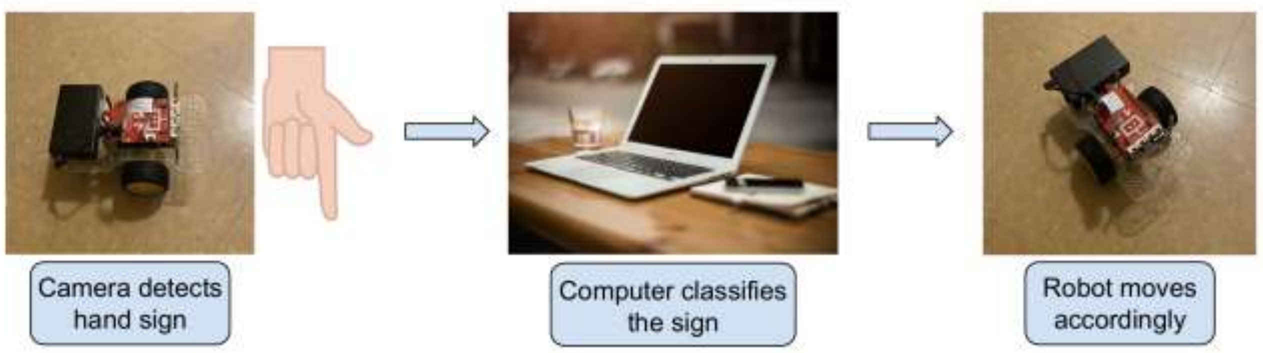
Sign Language Controlled Robot using Pre-Trained Models

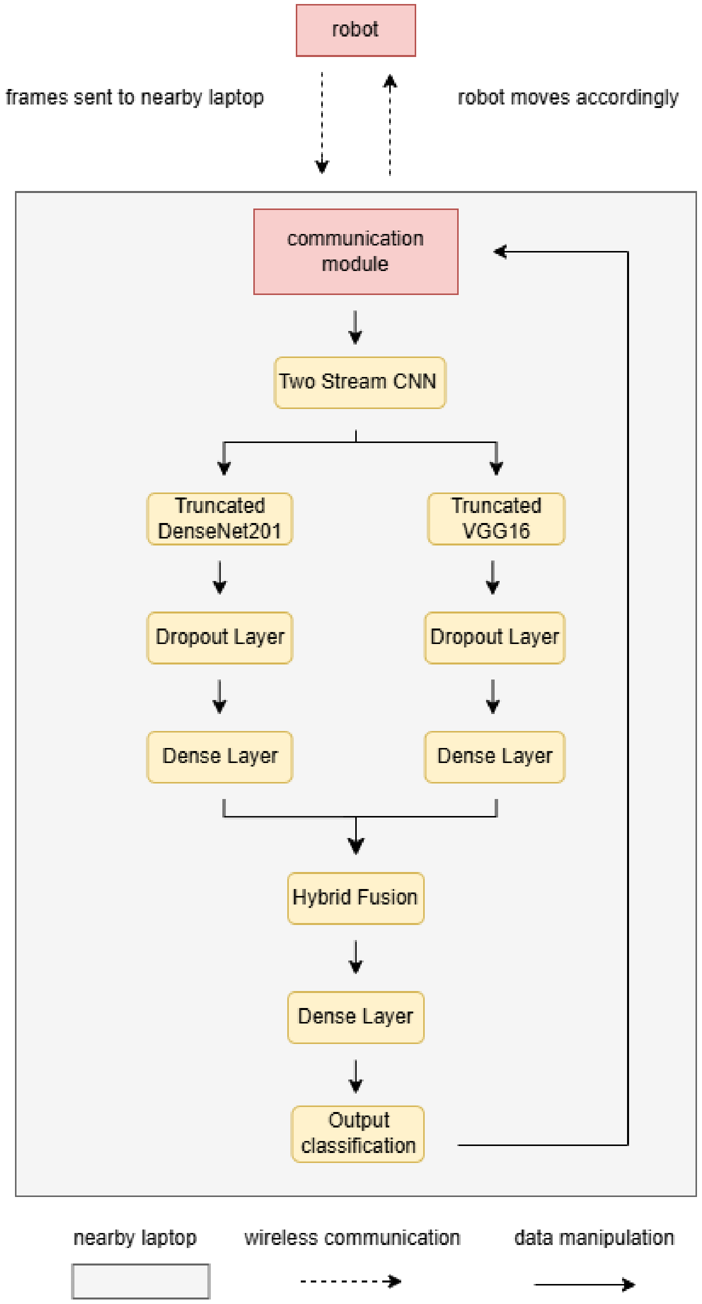
Introduction



- Project goal: build a robot that understands ASL gestures and moves accordingly
- Uses computer vision pre-trained models and a Raspberry Pi
- Aims to enhance accessibility and human-centered tech design

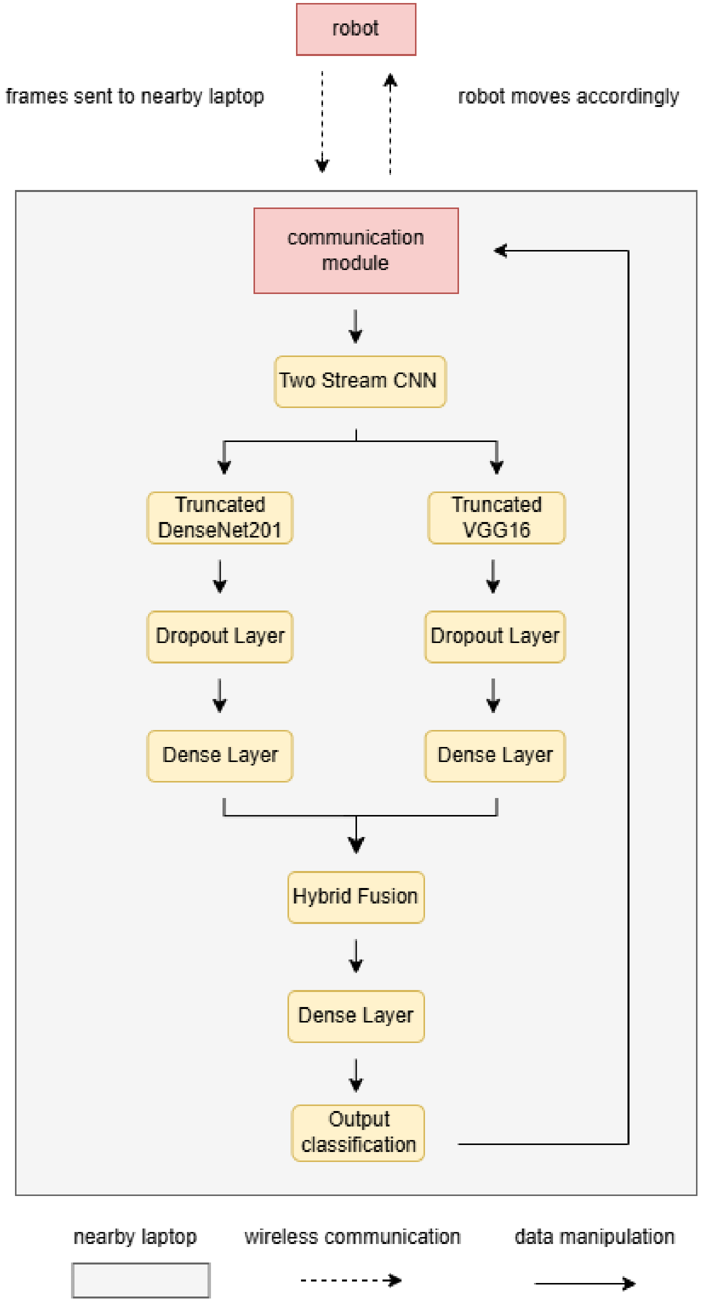
Data Architecture

- Laptop Robot Communication



Data Architecture

- Laptop Robot Communication
- Machine Learning Architecture using Pre-trained Models (training + validation)



Results

- Fusion model improved cross-dataset performance:78% on Kaggle (trained on My Images) 82% on My Images (trained on Kaggle)
- Average prediction time: 0.24s
- Average total response time: 2.63s
- Fusion model = better generalization + real-time capable

Conclusion

- Demonstrated strong accuracy across environments
- Real-time response makes it suitable for accessibility tools
- Promising step toward practical, inclusive HMI systems